Subject: ASD&D **Semester:** V

Class: TE **Batch:** T11

Name: Viraj Bhiwandkar **Roll No:** 24010014

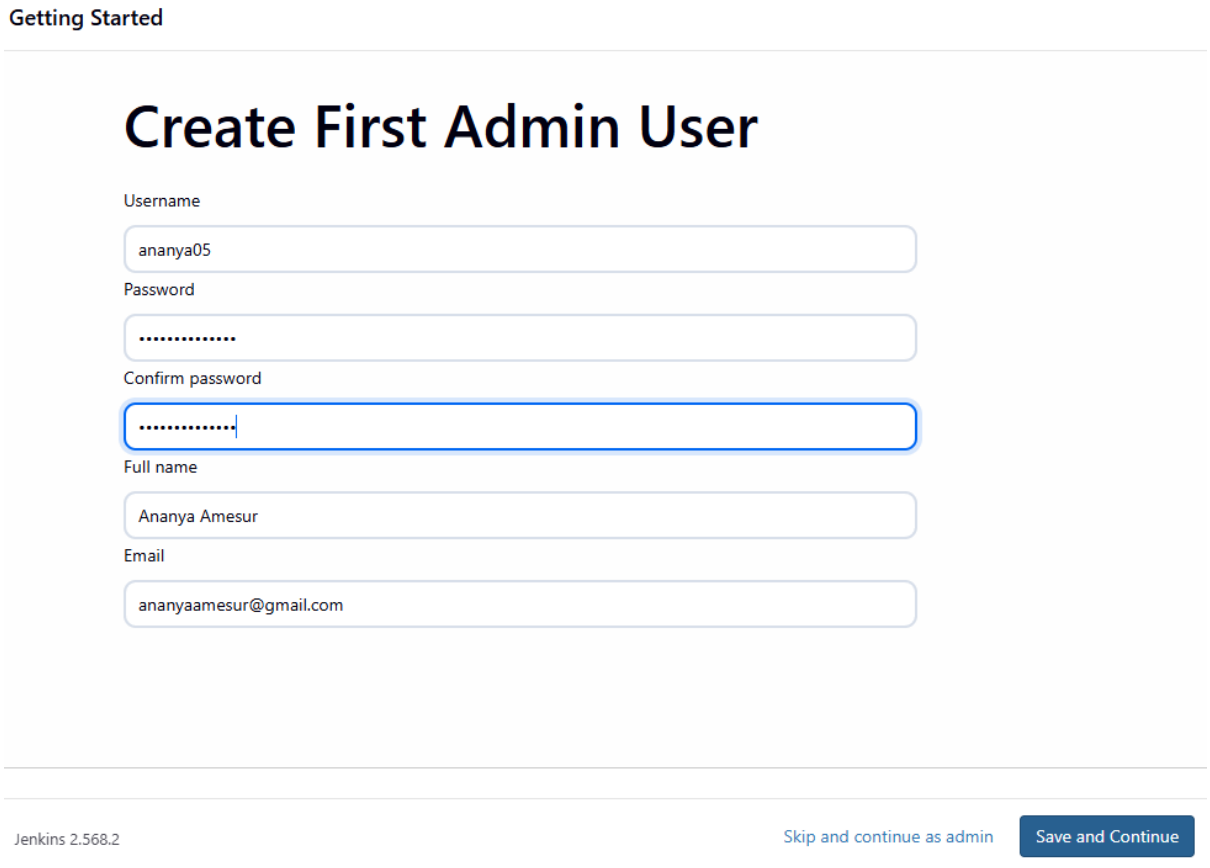
Experiment 5

Aim: To automate application deployment using Github/Jenkins.

Lab Outcome: Mapped LO3

LO3: Build and manage automated CI/CD pipelines for application development, testing, packaging and deployment.

Execution:



New Item

Enter an item name

GitHub-Jenkins-Demo

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

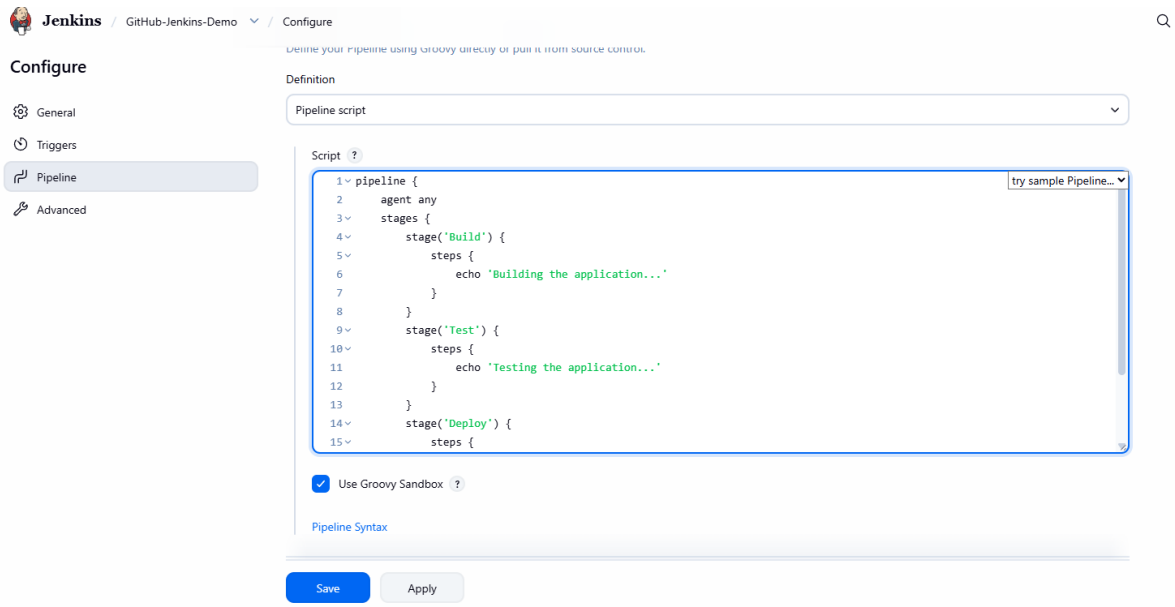
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK



GitHub-Jenkins-Demo / #1

ut

f1'

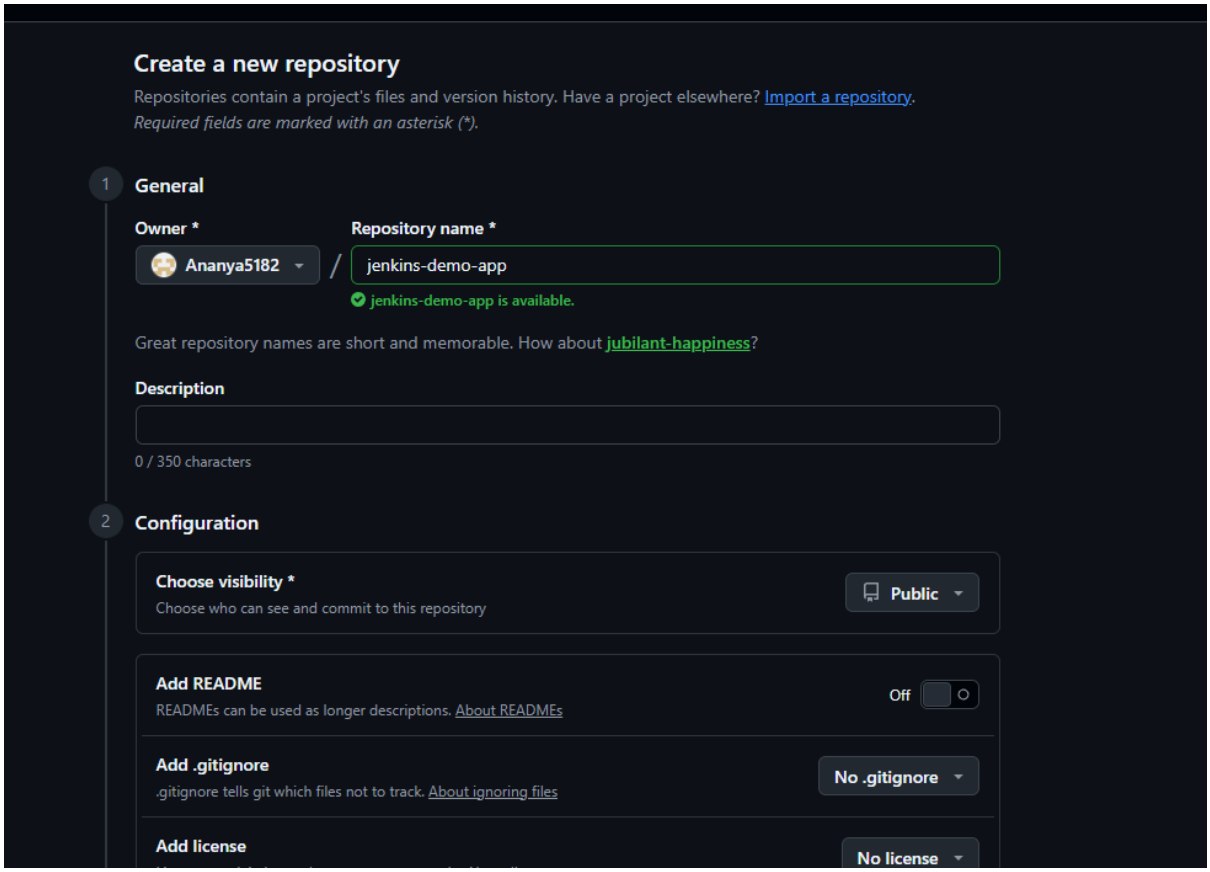
iew

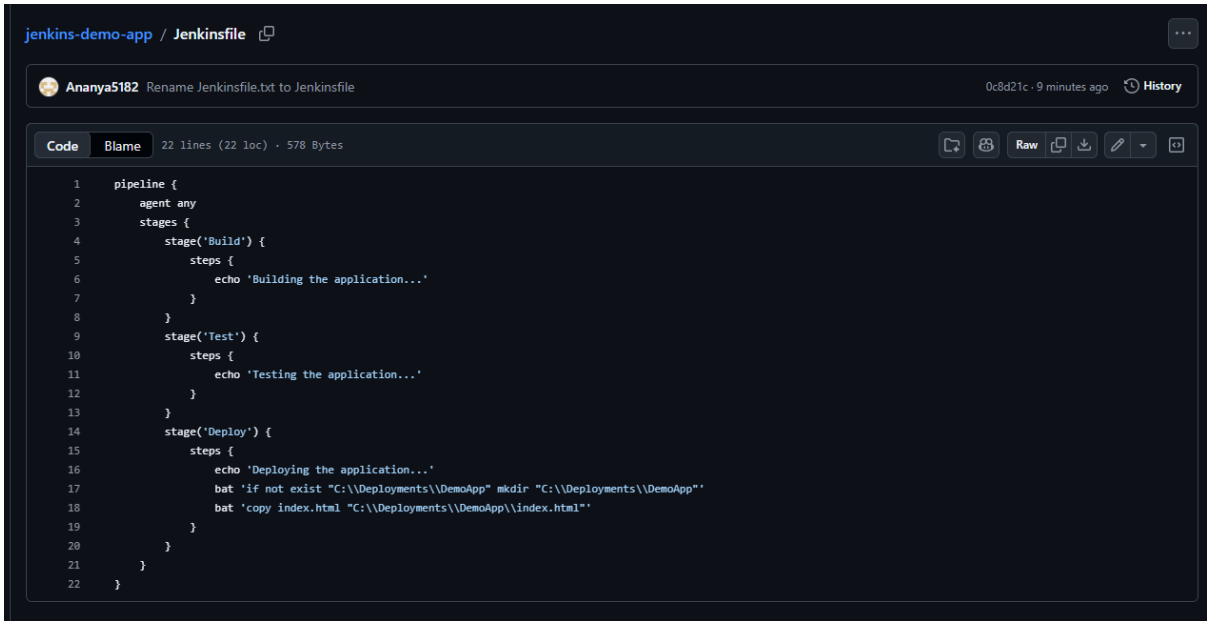
rmation

tage

Console

```
Started by user Ananya Amesur
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```


Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1
General
Owner *
Repository name *
Ananya5182
/
jenkins-demo-app
jenkins-demo-app is available.
Great repository names are short and memorable. How about jubilant-happiness?
Description
0 / 350 characters
2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
Off
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
No license


jenkins-demo-app / Jenkinsfile
Ananya5182 Rename Jenkinsfile.txt to Jenkinsfile
0c8d21c · 9 minutes ago
History
Code
Blame
22 lines (22 loc) · 578 Bytes
Raw
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Building the application...'
}
}
stage('Test') {
steps {
echo 'Testing the application...'
}
}
stage('Deploy') {
steps {
echo 'Deploying the application...'
bat 'if not exist "C:\\Deployments\\DemoApp" mkdir "C:\\Deployments\\DemoApp"'
bat 'copy index.html "C:\\Deployments\\DemoApp\\index.html"'
}
}
}
}

jenkins-demo-app / index.html in main

Cancel changes | Commit changes...

Edit | Preview

Spaces 4 No wrap

```
<!DOCTYPE html>
<html>
<head>
    <title>CI/CD Demo Application</title>
</head>
<body>
    <h1>Application Deployed Successfully via Jenkins!</h1>
</body>
</html>
```

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git ?

Repositories ?

Repository URL ?

https://github.com/Ananya5182/jenkins-demo-app.git

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

+ Add Branch

Repository browser ?

(Auto)

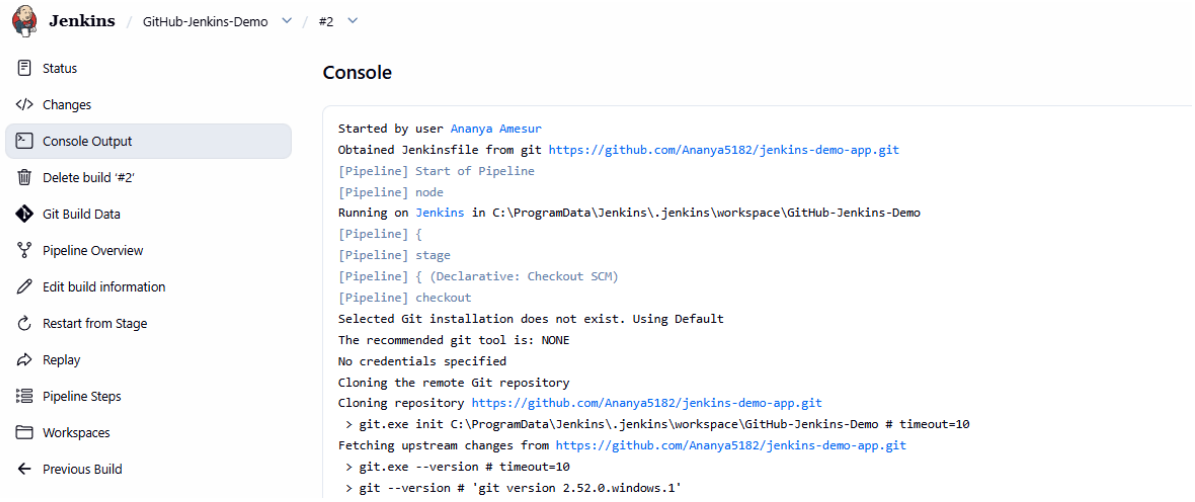
Additional Behaviours

+ Add

Script Path ?

Jenkinsfile

Lightweight checkout ?




```
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/Ananya5182/jenkins-demo-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe config remote.origin.url https://github.com/Ananya5182/jenkins-demo-app.git # timeout=10
 > git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 0c8d21cde227e66276fd9b2df19fc1815a852684 (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 0c8d21cde227e66276fd9b2df19fc1815a852684 # timeout=10
Commit message: "Rename Jenkinsfile.txt to Jenkinsfile"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the application...
```

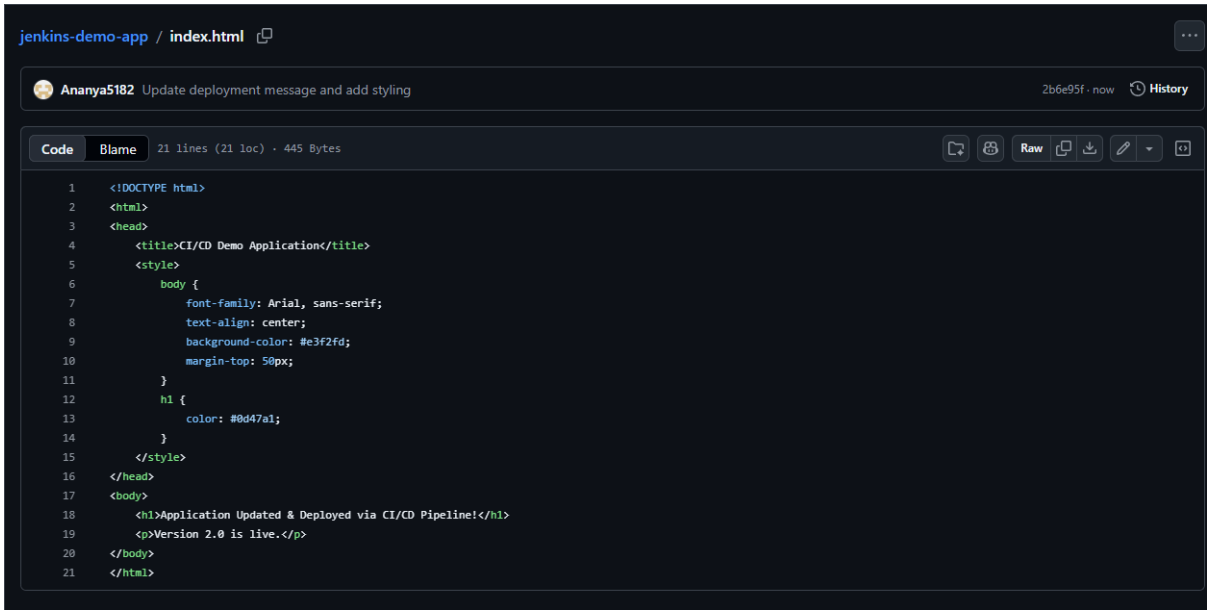
```
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo>if not exist "C:\Deployments\DemoApp" mkdir "C:\Deployments\DemoApp"
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo>copy index.html "C:\Deployments\DemoApp\index.html"
        1 file(s) copied.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



Application Deployed Successfully via Jenkins!




```html
<!DOCTYPE html>
<html>
<head>
    <title>CI/CD Demo Application</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            background-color: #e3f2fd;
            margin-top: 50px;
        }
        h1 {
            color: #0d47a1;
        }
    </style>
</head>
<body>
    <h1>Application Updated & Deployed via CI/CD Pipeline!</h1>
    <p>Version 2.0 is live.</p>
</body>
</html>
```

GitHub-Jenkins-Demo / #3

```
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo>if not exist "C:\Deployments\DemoApp" mkdir "C:\Deployments\DemoApp"
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo>copy index.html "C:\Deployments\DemoApp\index.html"
        1 file(s) copied.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

